

Anonymity: Each user can interact with the blockchain with a generated address, which does not reveal the real identity of the user.

Security features of blockchain

Blockchain helps in recording the transactions of any digital assets exchanged between two unknown parties. The security aspects it supports are critical in ensuring transparency, confidentiality, and protection against fraud. The following are the high-level security features of the blockchain.

Ledger: Ledger records every transaction in the blockchain. It is a chain of blocks, and information in the block is immutable. The distribution of ledger is done for all the participating nodes.

Chain of blocks: Blockchain is a chain of blocks. Each block has the hash value of the previous block and this forms a chain. Correction to data in a block (say n) will change the hash value and will not validate the hash stored in the next block ($n+1$). This is a chain reaction that affects the overall chain. Therefore, this increases the protection of sensitive data or information.

Confidentiality: Blockchain provides confidentiality by enabling users of a ledger to see authorised transactions only.

Transparency: Blockchain shares the ledger with all nodes and uses consensus algorithms to reach a consensus among them. Consensus algorithms also ensure the ordering and execution of transactions.

Cryptology: Cryptology enables secure transactions and makes blockchain immutable by using hash based algorithms, which produce a fixed hash based on the content of the block.

Smart contracts: Smart contract is a computer code running on top of a blockchain. It contains a set of rules under which the parties agree to interact with each other. If the predefined rules are fulfilled, then the agreement is automatically executed. No contract will execute without the network consensus.

Blockchain reference architecture

Figure 1 shows the logical application architecture of a blockchain system with its key components and layers. While there exist many standard logical architectures for the blockchain, the detailed blockchain logical reference architecture shown here is based on my practical experience across various domains and technologies.

The various layers for an enterprise application are classified as:

- Access layer
- Channel layer
- Customer experience layer
- Consumer layer
- Business and information services layer
- Information management and analytics service layer
- Blockchain services layer
- Enterprise integration layer
- Enterprise systems layer
- Distributed ledger stacks
- Security, and identity and access management layer
- Application and infrastructure management

Access layer: Various stakeholders, both internal and external, are part of this layer. They are the primary users of the systems.

Channel layer: Stakeholders use channels to interact with enterprises. They engage with various departments or business units of the enterprise over multiple channels, both physical and digital.

Customer experience layer: This layer is a collection of presentation components and services. It also constitutes the critical systems that are in use by enterprises and business units to engage stakeholders.

Consumer layer: The consumer layer is a gateway for channels and aggregates functionalities through composition and orchestration, mediation, and routing.

Its key capabilities are:

- **API gateway:** A single point of

entry for consumers to access back-end services. The service composition and orchestration is based on customer journey and context. This capability is provided by API management platforms.

- **State management:** Manages state and transition. Control logic is decoupled from the user interface and managed at the consumer end.

Business and information services layer: This layer is designed using ‘smart architecture’ principles, which provide cross-channel capabilities.

Information management and analytics services layer: This layer focuses on data ingestion, processing, complex event processing and real-time analytics and insights. A basic service system provides fundamental data services, which include data access, data processing, data fusion, data storage, identity resolution, geographic information service, user management, and inventory management.

Blockchain services layer: This layer acts as a gateway for reaching out to the blockchain infrastructure. It consists of various services related to identity management, key management, cryptography, machine learning (ML) and business intelligence (BI). Identity, key, and cryptographic services are critical for data integrity. BI and ML are used for analysis and reporting purposes.

Enterprise integration layer: This layer is a key enabler as it provides mediation, transformation, protocol and routing capabilities, and acts as a gateway to integrate with core enterprise systems. It also provides aggregation and broker communications. The enterprise integration and presentation layers are responsible for the integration of the blockchain infrastructure with the enterprise application stack.

Enterprise systems layer: The enterprise systems layer represents a collection of enterprise back office core systems and applications that house business data.